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in Arshath



Profile

MSc Cyber Security student specializing in Threat & Risk Analysis (TRA), with experience in threat modeling (STRIDE), vulnerability assessment, and secure system design. Currently evaluating TRA quality for industrial (OT) systems at Siemens. Skilled in analyzing attack surfaces, identifying security risks, and validating controls across applications and infrastructure.

Education

- 03/2025–09/2026 **Master of Science (MSc) Cyber Security**
Hochschule der Bayerischen Wirtschaft gGmbH (HDBW), Munich, Germany. (Ongoing, Current Grade: 2.0 / 5.0)
- Created and analyzed system and data-flow representations to support structured threat modeling and risk analysis using STRIDE and Microsoft Threat Modeling Tool.
 - Developed a Python-based ML system for cyberattack detection using network traffic data, achieving **92%** accuracy through feature engineering and model optimization. Project Link: [/Classifying-CyberAttacks-Using-ML](#)
 - Applied secure-coding principles while analyzing vulnerable implementations (OWASP Top 10), focusing on how subtle coding errors can undermine cryptographic security.
 - Audited and hardened Linux and Windows systems through log analysis, authentication controls, user management, and ACL enforcement.
 - Analyzed GDPR, NIS2, EU AI Act, and security standards to identify compliance requirements, data protection risks and mitigation gaps.
- 06/2019–04/2023 **Bachelor of Engineering (B.Eng.) Computer Science**
Anna University, Chennai, India. (Graduate - Indian CGPA: 8.38 / 10.0)
- **Bachelor Thesis:** AI-Enhanced Eye Tracking System for Tetraplegia Patients
 - Developed and led a deep learning-based eye-gaze communication system for tetraplegic patients using real-time gaze tracking, enabling message transmission via Twilio through optimized dataset collection and model tuning.
 - **Paper Publication Link:** [/AI-Enhanced-Eye-Tracking-System-For-Tetraplegia-Patients](#)
 - **Mini Project:** NALA: The Virtual Voice Assistant using Python
 - Developed a Python-based voice-controlled system using SpeechRecognition and pyautogui to execute tasks through intuitive voice commands.
 - Github link : [/NALA-The-Virtual-Voice-Assistant](#)
 - **IBM Project:** Vehicle Performance Analyzer using Machine Learning
 - Developed a vehicle performance analyzer on IBM Cloud using front-end tech and ML to predict vehicle mileage.
 - Github link : [/Vehicle-Performance-Analyzer](#)

Experience

- 04/2026–present **Master's Thesis (Cybersecurity), Siemens AG, Munich, Germany**
- Researching methods to evaluate completeness and consistency of Threat & Risk Analysis (TRA) for industrial (OT) systems using measurable metrics
 - Translating expert review practices into rule-based approaches to assess completeness and consistency of TRA.
 - Basic understanding of industrial control systems (ICS) and OT security concepts, familiarity with IEC 62443 principles.
 - Exploring automation of TRA quality checks using structured data (e.g., JSON) and collaboration with domain experts.
- Tool/skills - Threat and Risk Analysis, Security Analysis, JSON, Data Analysis, Research Methodology**

- 07/2024–01/2025

Programmer Analyst Trainee, *Cognizant Technology Solutions India Pvt. Ltd.*, Chennai, India.
 - Validated security-sensitive workflows (authentication, input validation, APIs), identifying logic flaws and security weaknesses relevant to secure software design.
 - Developed and maintained Cypress automated test suites, improving defect detection by **25%**; validated REST APIs using Postman including authentication flows and HTTP behavior.
 - Integrated automated tests into Git-based CI/CD pipelines for reliable releases and delivered internal knowledge sessions on test automation workflows.**Tool - Agile, JIRA, Cypress, REST APIs, Git-Lab**
- 12/2023–06/2024

Programmer Analyst Trainee - Intern, *Cognizant Technology Solutions India Pvt. Ltd.*, Chennai, India.
 - Supported development and testing of a Java-based retail management application in a team environment.
 - Tested authentication flows, input validation, and core business logic across web and API layers.**Tool - Core Java, Spring Boot, MySQL, Selenium**
- 10/2023–11/2023

Software Developer Intern, *Digitus Business Solutions Pvt Ltd*, Bangalore, India.
 - Developed responsive enterprise front-end components using Vue.js and Vuetify, enhancing usability and optimizing user and purchase-order workflows.
 - Implemented secure backend features with Spring Security, including JWT-based authentication and role-based access control.**Tool - Vue.js, Vuetify, Java Core, Spring Boot, Postman, Eclipse, Git**

Personal Cybersecurity Projects And Labs

- GitHub

Security labs & projects, pentesting practice, Python automation github.com/Arshathhaq

Blog

Hands-on write-ups on Security Labs & Projects medium.com/@Arshathhaq97

Security Labs

 - o **SIEM & Incident Response Lab**: Built SOC lab using Wazuh & TheHive to detect simulated attacks, perform alert triage, and analyze logs for threat detection (applicable to IT/OT monitoring).
 - o **Active Directory Security Lab**: Deployed AD environment and analyzed privilege escalation scenarios to understand identity-based attack paths and associated risks.
 - o **Web & API Security Lab**: Evaluated OWASP Top 10 vulnerabilities, validating security controls and analyzing impact from a risk assessment perspective. **Tools: Burp Suite.**
- Participations

Active on TryHackMe (*HMA.1002*) and MetaCTF, focusing on attack detection, log analysis, and web exploitation.

Skills

- Technical Skills

 - o **Security & TRA**: Threat modeling, vulnerability assessment, risk assessment and control evaluation, web/API security testing (auth, input validation, access control).
 - o **Tools**: Wazuh (SIEM), Burp Suite, Wireshark, Nmap, Metasploit, Postman.
 - o **Programming**: Python, Java, Bash, JavaScript; basic Go; familiarity with C/C++.
 - o **Systems**: Linux, Windows, Active Directory; Docker, VirtualBox.
- Core Skills

Analytical thinking, technical documentation, collaborative problem solving

Languages

- English

CEFR C1

Native / Bilingual Proficiency
- German

CEFR A2 (improving)

Elementary Proficiency